TRACK A / TECHNICAL REPORT

Lumen
Coronary Evidence Explorer

Connect each coronary prediction to named anatomy and exact model reasoning. Challenge the demonstration with the complete development cohort, including false positives, missed disease and incorrect flags that persist across five input profiles.

FINAL HOLDOUT EVALUATED
Development and final test evidence are reported separately. All claims remain limited to the evaluated source dataset.

242 records

Every development row

No curated success selection

20 model views

5 profiles x 4 targets

Exclude the entire outer fold

One audit trail

Find, inspect, reveal

Recorded outcomes disclosed

Five profiles can agree and still be wrong.

LCX: 60 incorrect flags among 172 agreeing development records. Agreement is not confidence.

Figure 1. Development audit design and its recorded counterexample. This is a static evidence diagram, not a browser screenshot. Every model view uses an excluding-fold model.

Clinical targets	Data partition	Deployment compute
CAD + LAD + LCX + RCA	242 development 61 final holdout	CPU experiments + inference Portable browser runtime

Contribution. A reproducible audit connects all 242 development records to their excluding-fold models, five-profile comparisons, generic named arteries and exact explanations. Outcome-aware filters expose failures beyond the eight introductory examples. Source-bound receipts support inspection; this is an engineering contribution, not a claim of a new diagnostic algorithm.

Educational research prototype. Not validated for clinical use or a substitute for diagnostic imaging. Probabilities do not describe stenosis severity or patient-specific plaque locations.

01 / DATA AND VALIDATION

Keep outcomes out of the inputs

303 records
Source hash checked

242 development
Nested model selection

61 final holdout
Scored once after freeze

All four target columns are excluded from every input pipeline.

Source and targets. The UCI extension of Z-Alizadeh Sani contains 303 records [1]. Cath supplies the overall CAD label; LAD, LCX and RCA supply vessel status. All four angiography columns are excluded from every task, leaving 55 predictors. Source hashes and frozen row identities are saved.

Preprocessing. Numeric median imputation and scaling, plus categorical imputation and one-hot encoding, are refitted within each training partition. Inference preserves the fitted values and reports unseen categories. Original source spelling is retained in the encoder.

Input group	Count	Operational availability meaning
History	17	Demographics and recorded history indicators
Examination	14	Recorded symptoms and examination findings
ECG	7	Recorded ECG indicators
Laboratory	14	Laboratory measurements, source encodings retained
Echo	3	EF-TTE, Region RWMA and VHD

The five availability groups are an explicit experiment definition informed by the source categories. Laboratory and echo are split operationally; the dataset does not establish a temporal care pathway. Verified units are used only where supported by source evidence [1,2].

Nested selection (V2). Five shared CAD-stratified outer folds evaluate the entire selection procedure. Three target-stratified inner folds select C (0.01, 0.1, 1, 10), unweighted/balanced logistic fitting, and optional sigmoid calibration by pooled inner log loss. Calibrated candidates use further cross-fitted training scores; outer validation records are excluded from fitting and calibration [4,5].

Operating rule. The selected candidate maximizes inner out-of-fold F1 over thresholds 0.05-0.95 at 0.01 steps. Ties favor proximity to 0.5, then the higher point. Outer classification metrics use the threshold fitted without each validation record. AUC and Brier use probabilities, independent of this classification threshold [6].

Development audit: 0 missing cells; 0 duplicated input rows; 1 CAD/vessel label discrepancy preserved. V2 followed inspection of V1 development results, so development evidence includes experiment-selection uncertainty.

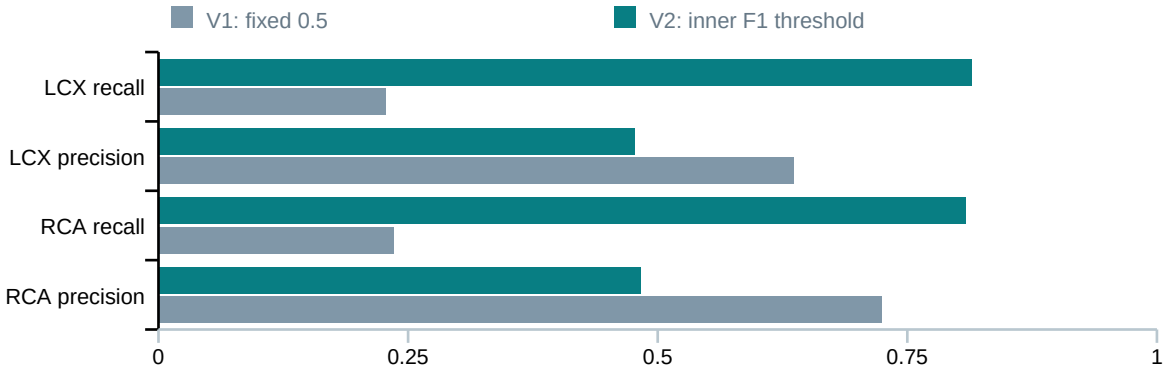
02 / PREDICTIVE EVIDENCE

Detection improves with a cost

V2 pooled development out-of-fold results, n=242. Each record is predicted by an excluding outer model. Intervals bootstrap fixed OOF predictions (1,000 resamples); they omit refitting and experiment-selection uncertainty.

Target	AUC	95% AUC interval	Acc.	Prec.	Recall	F1	Brier
CAD	0.926	0.889 - 0.956	0.868	0.881	0.942	0.911	0.096
LAD	0.814	0.759 - 0.864	0.731	0.718	0.887	0.794	0.174
LCX	0.701	0.632 - 0.765	0.591	0.478	0.815	0.602	0.211
RCA	0.722	0.653 - 0.783	0.612	0.483	0.809	0.605	0.202

Figure 2. Lower fitted operating thresholds increase LCX/RCA recall while reducing precision. This is a classification tradeoff; vessel discrimination does not improve over the initial baseline.



V2 selected rule	True positive	False negative	False positive	True negative
LCX	75	17	82	68
RCA	72	17	77	76

The initial fixed-0.5 V1 baseline had LCX/RCA recall 0.228/0.236 and precision 0.636/0.724. V2 raises recall to 0.815/0.809 while precision falls to 0.478/0.483. The additional false positives remain visible, rather than being hidden behind a higher F1.

Probability quality. Brier and log loss are reported alongside ranking and classification metrics. A sigmoid step does not establish calibrated probabilities in a new population; Brier mixes reliability and discrimination, and the saved reliability bins are small [5].

These are development results, not external validation or medical operating thresholds. The four independent models do not impose a coherent joint CAD/vessel distribution.

03 / INPUT PROFILES AND COHORT AUDIT

Agreement must survive a failure audit

Five cumulative input profiles are fitted separately under the same frozen development identities and outer folds. The full-input profile reuses V2. This evaluates model evidence with available groups; it does not estimate medical test utility, cost or causal benefit.

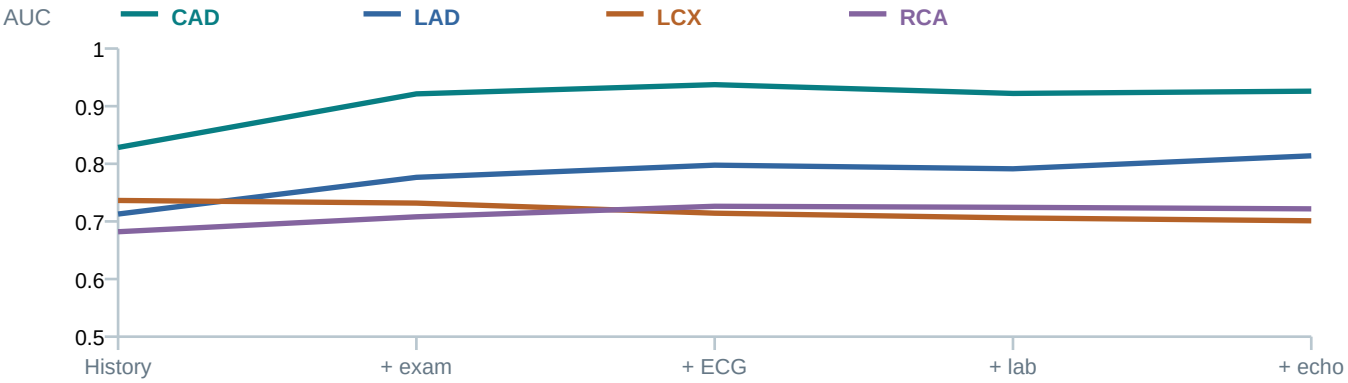


Figure 3. Development ROC-AUC by cumulative input profile. Each profile has its own nested fitting, calibration selection and threshold rule; groups are not merely nulled in the full-input model.

All 242 records, including stable incorrect flags

Target	Five profiles agree / 242	Incorrect among agreeing	FP / FN
CAD	182	12	10 / 2
LAD	161	33	32 / 1
LCX	172	60	51 / 9
RCA	164	58	56 / 2

Each flag uses its model's already fitted research threshold. Agreement selects a subset; it does not improve a retained prediction. LCX remains wrong on 60/172 agreeing records, and RCA on 58/164. These overlapping models are not independent witnesses.

The audit filters by target, outcome, profile agreement, outer fold and a shared fixed/research decision rule. Every selected record opens the anatomy and exact score. Recorded outcomes define filters: this is descriptive development review, not a blind or prospective test.

Bounded model comparisons remain exploratory

V1 and V3 retained prior/tree baselines. A later 200-fit nested CatBoost CPU probe took 190.33s and 423MB peak RAM: LAD AUC improved from 0.814 to 0.829, while CAD/RCA weakened and LCX was unchanged. These mixed development results did not replace the frozen V2 procedure or reopen final evaluation.

Measured elapsed time in the local CPU runs: V1 14.4s, V2 39.6s, V3 19.9s, availability 121.1s. V2 timing includes bounded search, export checks, conditional bootstraps and artifact creation. No GPU was used.

Model-family and input-profile decisions were frozen before final holdout scoring. Final outcomes were not used to retune the submitted procedure.

04 / ANATOMY AND INTEGRATION

One mapping from score to vessel

Browser
3D meshes + inputs

—

Python API
Schema + case routing

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JSON artifacts
Score + explanation

Truth reveal is a separate API path. Edited inputs have no observed edited outcome.

Reference geometry. Official BodyParts3D memberships define heart FMA7088, LAD FMA3862, LCX FMA3895 and RCA FMA50039 [3]. Source OBJ parts are merged, triangulated and transformed into selectable Three.js groups [7]. Provenance retains hashes, memberships and modifications.

Visual semantics. The target key selects matching geometry, probability and explanation. Overview, opposite view, mesh-bound focus and artery isolation complement rotate/zoom and the surface toggle. Isolation hides other groups and restores the chosen surface state. Probability colors the whole vessel; no patient geometry or lesion coordinates are inferred.

Grouped interventional linear SHAP (log-odds)

For transformed feature values z, training reference means mu, coefficients beta and intercept b, the reference and contributions are:

```
reference = b + sum(beta[j] * mu[j])
contribution[j] = beta[j] * (z[j] - mu[j])
score = reference + sum(contribution[j])
probability = sigmoid(score)
```

Encoded terms are grouped into raw inputs. Calibration slope/offset are composed into coefficients/intercept. The complete waterfall shows the reference, six largest terms, all remaining terms as a residual, and the final score. Sigmoid applies only at the endpoint; sigmoid(reference) is not mean predicted probability.

Actual SHAP LinearExplainer with an Independent masker verifies this decomposition [9], using every encoded training-fold row. Independent masks break feature dependencies and may form invalid intermediate one-hot states. Grouped terms are interventional model log-odds values, not conditional explanations or percentage-point probability changes.

Correlated clinical inputs can receive different credit under a correlation-aware explanation. The displayed terms are model associations, not treatment effects. Input edits explore fitted-model sensitivity and do not simulate a clinical intervention.

Verification and interaction

The new compact export matched all 4,840 saved development probabilities exactly. It preserves the original eight introductory cases and adds the remaining development records. Detailed contributions are calculated lazily from the same 100 profile/target/fold models; labels never enter inference.

The separate complete-cohort actual-SHAP certificate passed 4,840 explanations across 100 models; maximum grouped-term error 5.55e-17 and reconstructed score error 5.33e-15, tolerance 1e-12. Its Independent background excludes the record, the entire outer fold and every final holdout row. The original 160-explanation certificate remains scoped to the eight-case roster. These checks verify the specified decomposition, not clinical explanation utility.

Audit selections open the same anatomy, score waterfall and five-profile matrix. Explicit label reveal is separate from prediction routes; outcome-aware filter selection is disclosed. Input edits disable matching recorded-outcome comparison. Unlabelled new inputs use the frozen full-development models. UCI411-288 is an existing LCX missed-disease example with five agreeing negative research flags.

05 / FINAL EVIDENCE AND REPRODUCIBILITY

Evidence boundaries and release record

Final holdout: n=61. The primary full-input procedure was frozen before scoring once. No holdout fitting, calibration, threshold or profile selection was performed.

Target	AUC	95% AUC interval	Acc.	Prec.	Recall	F1	Brier
CAD	0.908	0.815 - 0.972	0.836	0.851	0.930	0.889	0.110
LAD	0.893	0.804 - 0.966	0.852	0.814	0.972	0.886	0.126
LCX	0.748	0.621 - 0.859	0.607	0.535	0.852	0.657	0.216
RCA	0.724	0.579 - 0.844	0.590	0.500	0.760	0.603	0.209

Positive/negative counts: CAD 43/18; LAD 36/25; LCX 27/34; RCA 25/36. Fitted thresholds: CAD 0.21, LAD 0.36, LCX 0.30, RCA 0.25. Conditional intervals use 1,000 fixed-prediction patient resamples; no refitting uncertainty.

LCX: 23 TP, 4 FN, 20 FP, 14 TN. **RCA:** 19 TP, 6 FN, 19 FP, 17 TN.

Limits

Small dataset; no external or prospective validation. LCX/RCA discrimination is limited, with a large false-positive cost. Conditional intervals omit refitting uncertainty. Independent targets can disagree. Reference anatomy cannot locate patient plaque; edited inputs lack outcome labels. Units are used only where verified.

Reproduce and verify

Start with `npm ci`, `python3 server.py`; open `http://127.0.0.1:8766/`. Python standard library performs inference; the static build evaluates entered inputs in browser memory. Ordinary browser graphics render WebGL. `docs/cohort-audit.md` documents audit use, export and pinned CPU verification commands.

Credits and AI assistance

Dataset: Alizadehsani, R., Roshanzamir, M., & Sani, Z. (2013), UCI 411, CC BY 4.0 [1]. BodyParts3D, © The Database Center for Life Science licensed under CC Attribution 4.0 International [3]. Geometry was merged, triangulated, translated and rotated. Three.js is MIT licensed [7].

OpenAI Codex assisted with research, implementation, experiments, debugging and documents. The human submitter owns verification and judging explanations. AI assistance is declared here and in README; Devpost Built With must include it [8].

Primary references

[1] UCI extension of Z-Alizadeh Sani. DOI: 10.24432/C5461K.
[2] Alizadehsani et al. Laboratory and echocardiography features (2013). DOI: 10.5812/cardiovascmed.10888.
[3] BodyParts3D archive: license and source download.
[4] scikit-learn: nested cross-validation.
[5] scikit-learn: probability calibration.
[6] scikit-learn: tuning the decision threshold.
[7] Three.js documentation.
[8] Multimodal AI Hackathon 2026: rules; supplied Track A and submission guidelines.
[9] SHAP LinearExplainer: interventional linear attribution.